

Homework 2

due 25.10.2023, 14:00

Exercise 2.1 - Collective Communication (10 points)

On stromboli in the folder

/archive/keep/Lab2/ex2/

you can find 64 files `data_0.dat`, ..., `data_63.dat`. Each file contains 100 numbers. Write a program with C and MPI that computes and prints out the mean, variance, minimum and maximum of the data in each file, and of all the data together.

- A process with rank `rank` should process data files `data_n.dat`, where $n \bmod \text{numprocs} = \text{rank}$.
- Every file should be read only once.
- No arrays should be used for storing the data. The necessary calculations can be carried out “on the fly”.
- For data $d_i, i = 0, \dots, N - 1$, use the estimators

$$\begin{aligned}\text{mean}(d) &= \frac{1}{N} \sum_{i=0}^{N-1} d_i \\ \text{var}(d) &= \frac{1}{N-1} \sum_{i=0}^{N-1} (d_i - \text{mean}(d))^2\end{aligned}$$

Hint: rewrite the second expression, so it can be computed in one pass over the data

- Execute and time your program with 1,2,4 and 64 cores on stromboli.